

# CodeGraph – Status

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| Field          | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Revision       | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Created        | 2026-05-21                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Last modified  | 2026-10-05T09:59:56Z                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Status         | active                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| Status summary | <p>Append-only ledger of every CodeGraph-related event across the constitution submodule's consuming projects: npm updates, sync runs, index regenerations, validate runs, HRDs opened/closed against CodeGraph. Per Universal §11.4.80 + §11.4.65 (multi-format export). Both <code>scripts/code-graph_update.sh</code> and <code>scripts/codegraph_sync.sh</code> append here automatically. R2 (2026-05-22): replaced project-specific §107 anchor references with the universal §11.4 covenant anchor to keep this submodule fully decoupled per §11.4.28. R3 (2026-10-05, currency check): back-filled the §11.4.275 EXTENSION history (both 2026-09-25 rounds — the host-adaptive V8 heap-budget fix + §11.4.275(D) benchmark run, and the heap-ceiling correction + CodeGraph MCP wiring + the <code>serve --mcp</code> stray-daemon forensic incident) that had landed in <code>Constitution.md</code> but was never appended to this ledger; see the <code>## 2026-10-05</code> entry below. A handful of earlier §107 legend entries (2026-09-29) pre-date R2's re-name and are left verbatim as the historical record they are.</p> |
| Issues         | none                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Issues summary | —                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Fixed          | (n/a — ledger doc)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Continuation   | sibling <code>Status_Summary.md</code> carries the operator-readable digest per §11.4.53.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

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## Cadence + automation

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Per §11.4.80, every consuming project MUST run `codegraph_update.sh` at least weekly and `codegraph_sync.sh` after every update + on every meaningful source-change cadence (varies by project). Cadence floor: weekly. Cadence ceiling: per-CI-run.

The scripts append entries below automatically; manual entries are also acceptable when an operator runs the commands directly.

## Event ledger

---

2 0 2 6 0 5 2 1 0 5 3 8 5 3

(events appended below by the automation; newest at the bottom)

codegraph update FAILED (\$ : Z -  
caught) bluff

- before: 0.6.8

2 0 2 6 0 5 2 1 0 5 4 0 0 1

- target: 0.8.0

- after: unknown (mismatch — npm exit 0 was bluffing)

2 5

codegraph rolled back to . . Z -  
(Node-incompatibility)

- before: 0.8.0 (installed but non-functional on Node 25.x — engine range + V8 WASM JIT bug)

- after: 0.6.8 (last known-working on Node 25.x)
- decision: pin codegraph to 0.6.8 until Node 25 V8 issue is resolved upstream  
OR operator installs Node 22 LTS alongside
- §11.4 trail: codegraph\_update.sh correctly caught the failed upgrade via the  
post-update version check — script working as designed

2 0 2 6 2 1 1 2 4 9 0 3 follow-up HRD: ENV-CODEGRAPH-NODE25 to track when 0.7+/0.8+ becomes  
safe again

1 1 4

codegraph update FAILED (\$ . bluff caught)

- before: 0.6.8
- target: 0.8.0
- after: unknown (mismatch — npm exit 0 was bluffing)

codegraph version check

- current: 0.8.0
- latest: 0.8.0
- action: no-op (already at latest)

codegraph version check

- current: 0.8.0
- latest: 0.8.0
- action: no-op (already at latest)

## codegraph version check

- current: 0.8.0
- latest: 0.8.0<sub>2 0 2 6 0 5 2 1 1 5 1 7 0 7</sub>
- action: **no-op** (already at latest)

## codegraph version check

- current: 0.8.0
- latest: 0.8.0<sub>2 0 2 6 0 5 2 1 1 5 2 1 5 5</sub>
- action: **no-op** (already at latest)

## codegraph version check

- current: 0.8.0
- latest: 0.8.0<sub>2 0 2 6 0 5 2 1 1 5 2 6 3 4</sub>
- action: **no-op** (already at latest)

## codegraph version check

- current: 0.8.0
- latest: 0.8.0<sub>2 0 2 6 0 5 2 2 0 8 4 4 3 6</sub>
- action: **no-op** (already at latest)

## codegraph version check

- current: 0.9.2

2 0 2 6 0 5 2 8 1 0 2 1 0 8

- latest: 0.9.2

- action: **no-op** (already at latest)

– – T : : Z –

## codegraph updated

---

- before: 0.9.4

2 0 2 6 0 5 2 8 1 4 2 6 3 5

- after: 0.9.6

- npm command: `npm install -g @colbymchenry/codegraph@0.9.6`

– – T : : Z –

## codegraph version check

---

- current: 0.9.6

2 0 2 6 0 5 2 9 1 8 5 1 5 5

- latest: 0.9.6

- action: **no-op** (already at latest)

– – T : : Z –

## codegraph version check

---

- current: 0.9.7

2 0 2 6 0 6 1 9 0 6 4 8 3 4

- latest: 0.9.7

- action: **no-op** (already at latest)

– – T : : Z –

## codegraph version check

---

- current: 1.0.1

- latest: 1.0.1

- action: **no-op** (already at latest)

## codegraph version check

- current: 1.0.1
- latest: 1.0.1
- action: **no-op** (already at latest)

## codegraph version check

- current: 1.0.1
- latest: 1.0.1
- action: **no-op** (already at latest)

## codegraph version check

- current: 1.0.1
- latest: 1.0.1
- action: **no-op** (already at latest)

## codegraph version check

- current: 1.0.1
- latest: 1.0.1
- action: **no-op** (already at latest)

## codegraph updated

- before: 1.0.1

• after: 1.1.2

• npm command: `npm install -g @colbymchenry/codegraph@1.1.2`

## codegraph version check

• current: 1.1.2

• latest: 1.1.2

• action: **no-op** (already at latest)

## codegraph version check

• current: 1.1.2

• latest: 1.1.2

• action: **no-op** (already at latest)

## codegraph version check

• current: 1.1.3

• latest: 1.1.3

• action: **no-op** (already at latest)

## codegraph version check

• current: 1.1.3

• latest: 1.1.3

• action: **no-op** (already at latest)

## codegraph updated

- before: 1.1.6
- after: 1.2.0
- npm command: `npm install -g @colbymchenry/codegraph@1.2.0`

## codegraph version check

- current: 1.2.0
- latest: 1.2.0
- action: **no-op** (already at latest)

## codegraph version check

- current: 1.2.0
- latest: 1.2.0
- action: **no-op** (already at latest)

## codegraph version check

- current: 1.2.0
- latest: 1.2.0
- action: **no-op** (already at latest)

## codegraph version check

- current: 1.2.0

• latest: 1.2.0<sub>2 0 2 6 0 7 0 5 0 6 2 8 4 9</sub>

- action: **no-op** (already at latest)

– – T : : Z –  
codegraph version check

---

• current: 1.2.0

• latest: 1.2.0<sub>2 0 2 6 0 7 0 5 0 6 4 6 1 5</sub>

- action: **no-op** (already at latest)

– – T : : Z –  
codegraph version check

---

• current: 1.2.0

• latest: 1.2.0<sub>2 0 2 6 0 7 0 5 0 6 5 5 5 3</sub>

- action: **no-op** (already at latest)

– – T : : Z –  
codegraph version check

---

• current: 1.2.0

• latest: 1.2.0<sub>2 0 2 6 0 7 0 5 0 7 0 6 0 9</sub>

- action: **no-op** (already at latest)

– – T : : Z –  
codegraph version check

---

• current: 1.2.0

• latest: 1.2.0

- action: **no-op** (already at latest)

## codegraph version check

- current: 1.2.0
- latest: 1.2.0
- action: **no-op** (already at latest)

## codegraph version check

- current: 1.2.0
- latest: 1.2.0
- action: **no-op** (already at latest)

## codegraph version check

- current: 1.2.0
- latest: 1.2.0
- action: **no-op** (already at latest)

## codegraph updated

- before: 1.2.0
- after: 1.4.1
- npm command: `npm install -g @colbymchenry/codegraph@1.4.1`

## codegraph version check

- current: 1.4.1

• latest: 1.4.1<sub>2 0 2 6 0 8 1 3 0 7 1 9 4 1</sub>

- action: **no-op** (already at latest)

– – T : : Z –  
codegraph version check

---

• current: 1.5.0

• latest: 1.5.0<sub>2 0 2 6 0 8 1 3 0 7 4 7 0 9</sub>

- action: **no-op** (already at latest)

– – T : : Z –  
codegraph version check

---

• current: 1.5.0

• latest: 1.5.0<sub>2 0 2 6 0 8 1 3 0 8 0 5 0 1</sub>

- action: **no-op** (already at latest)

– – T : : Z –  
codegraph version check

---

• current: 1.5.0

• latest: 1.5.0<sub>2 0 2 6 0 8 3 1 1 4 3 9 0 9</sub>

- action: **no-op** (already at latest)

– – T : : Z –  
codegraph version check

---

• current: 1.6.0

• latest: 1.6.0

- action: **no-op** (already at latest)

## codegraph version check

- current: 1.6.0
- latest: 1.6.0<sub>2 0 2 6 0 9 0 1 1 2 3 4 3 9</sub>
- action: **no-op** (already at latest)

## codegraph version check

- current: 1.6.0
- latest: 1.6.0<sub>2 0 2 6 0 9 0 1 1 5 5 9 4 1</sub>
- action: **no-op** (already at latest)

## codegraph version check

- current: 1.6.0
- latest: 1.6.0<sub>2 0 2 6 0 9 0 1 1 7 1 8 5 9</sub>
- action: **no-op** (already at latest)

## codegraph version check

- current: 1.6.0
- latest: 1.6.0
- action: **no-op** (already at latest)

codegraph update FAILED (\$ Z -  
caught) bluff

- before: 1.6.0

- target: 1.6.1

- after: 1.6.0 (mismatch — npm exit 0 was bluffing)

codegraph update FAILED (\$ Z -  
caught) bluff

- before: 1.6.0

- target: 1.6.1

- after: 1.6.0 (mismatch — npm exit 0 was bluffing)

codegraph update FAILED (\$ Z -  
caught) bluff

- before: 1.6.0

- target: 1.6.1

- after: 1.6.0 (mismatch — npm exit 0 was bluffing)

codegraph update FAILED (\$ Z -  
caught) bluff

- before: 1.6.0

- target: 1.6.1

- after: 1.6.0 (mismatch — npm exit 0 was bluffing)

codegraph update FAILED (\$ bluff caught)

- before: 1.6.0

- target: 1.6.1

- after: 1.6.0 (mismatch — npm exit 0 was bluffing)

2026 09 25

EXTENSION history backfilled (currency check; events below occurred retroactively on discovery that this ledger had not recorded them)

Per constitution/Constitution.md §11.4.275's own two "EXTENSION landed 2026-09-25" blocks (cited verbatim below, not paraphrased from memory per §11.4.6):

- **§11.4.78 host-adaptive V8 heap budget + §11.4.275(D) benchmark run (round 1, 2026-09-25).** codegraph\_safe.sh 's single writer entry now computes its Node.js heap budget from measured MemAvailable instead of inheriting V8's ~4 GiB stock default, closing a real bulk-index OOM crash ( rc=134 ) on this repo's 584K-file/49M-edge tree — landed with 2 new regression tests + live /proc/<pid>/environ verification + a zero-regression full-suite comparison (commit 941841d3e7235a081e3199a4d3073b409d41753b ). The §11.4.275(D) deterministic benchmark ran via constitution/scripts/lumen/lumen\_verify.sh : a pre-declared 13-query golden set, N=3 byte-identical runs, recall 9/11 = 0.818, with the 2 failing/unsupported query classes ( .sh / .kt , unindexed by Lumen's chunker — confirmed via direct sqlite3 + Lumen's own source) reported honestly and tracked, never averaged away (commit ac532a21df78a447c2aceec30f63476348e93077 , docs/research/lumen\_benchmark\_20260925/RESULTS\_v2.md ).

- **§11.4.78 heap ceiling correction + CodeGraph MCP wiring (round 2, 2026-09-25).** The round-1 64 GiB heap cap is corrected — it was an INVENTED ceiling (~25% of this 251 GiB host's RAM), not a derivation of §12.6; fixed to `MemTotal * 60/100` computed fresh every run, the real unweakened §12.6 bound (commit `8a5f47ab2e0e7a05d1c1652ba5a3475751652062`, regression test U46). Separately: `.mcp.json` was found completely EMPTY — CodeGraph had ZERO agent-facing MCP presence in this repository despite its full query toolset. Fixed by registering `.mcp.json`'s `codegraph` server against a new, tested, §11.4.177-decoupled wrapper ( `constitution/scripts/codegraph/codegraph_mcp_serve.sh` ) that resolves the SAME validated patched runner `codegraph_safe.sh` (never bare stock-on-PATH). A SEPARATE, more serious forensic incident surfaced while building this: `serve - -mcp`'s file watcher WRITES (auto-syncs) the database and the server daemonizes past its own parent's death — a manual test spawned a stray daemon that collided with an in-flight bulk `sync`, crashing it with "database is locked" (~85 minutes of progress lost); recovered via a targeted `REINDEX` (3 rows, confirmed non-corrupting) + a bulk `sync` resume. Durable fix: the wrapper now ALWAYS passes `--no-watch`, proven by a real init'd scratch-project DB-mtime check (unchanged after a tracked-file edit with the server running). Six new regression tests landed ( `tests/test_codegraph_mcp_serve.sh` ), including the incident itself replayed as a repeatable test.
- **Honest boundary** (carried from the anchor text, §11.4.6): `.mcp.json` registration takes effect on a fresh session — the session that landed it could not self-restart to verify the MCP tools are end-to-end callable from inside it; that check remains a tracked §11.4.197 follow-up, not claimed complete here. Gate-code for §11.4.78/§11.4.275's recommended `CM-*` mechanism gates remains owed (§11.4.227(A); see `docs/owed_gate_implementations.md` OWED-GATE-096 through OWED-GATE-105).